



Enhancing multi-modal aspect-based sentiment classification via emotional semantic-aware cross-modal relation inference

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ABSTRACT

Multi-modal Aspect-based Sentiment Classification (MASC) determines the sentiment polarity of specific aspects in text-image pairs. Recent research has explored leveraging image-text relevance to improve MASC performance. However, existing approaches primarily focus on explicit alignments between textual aspects and visual objects or on the global relevance between entire texts and images, often overlooking the implicit emotional connections specific to aspects. In this work, we propose an aspect-level emotional cross-modal relation scheme that captures both explicit alignments and implicit emotional connections between text and image. Based on this scheme, we construct a new dataset, the Aspect-level Emotional Cross-modal Relevance dataset (AECR-Twitter), which contains 3,562 image-text pairs. We also introduce several methods for integrating cross-modal relevance into MASC. Experimental results across eight different model architectures consistently demonstrate the effectiveness of our aspect-level emotional cross-modal relation scheme in enhancing MASC performance, with F1 scores increasing by an average of 1.26% on Twitter-15 and 1.28% on Twitter-17. We release our data and code at <https://github.com/li9527yu/AECR-Twitter>.

1. Introduction

Multi-modal aspect-based sentiment classification (MASC) is a crucial fine-grained task in sentiment analysis. It focuses on determining the sentiment polarity (i.e., positive, negative, or neutral) of a text-image pair regarding a specific aspect mentioned in the text. For example, in Fig. 1, the goal of MASC is to identify the sentiment toward the given aspect, “Trey Songz” as positive. MASC has made significant progress in recent years (Ling, Yu, & Xia, 2022; Peng, Li, Wang, Zhang, & Zhao, 2024; Zhou et al., 2023). As social media platforms increasingly leverage multi-modal content for information sharing, accurate MASC is crucial for improving information comprehension and offers significant benefits for tasks such as opinion mining (Huang et al., 2025), recommendation (Zhang et al., 2024), and misinformation detection (Luvembe, Li, Li, Liu, & Xu, 2023).

Unlike text-based sentiment analysis, which considers only a single textual modality, multi-modal sentiment analysis requires integrating both textual and visual semantic information to determine sentiment polarity. Therefore, in multi-modal scenarios, the relationship between text and images plays a crucial role in sentiment classification. Relevant images can complement textual information, leading to a more comprehensive understanding of sentiment, whereas irrelevant images may introduce noise and mislead the model, resulting in incorrect sentiment predictions.

With the above consideration, recent research has explored improving sentiment analysis performance by leveraging the cross-modal relevance between images and texts. Ju et al. (2021) propose to leverage the cross-modal relation scheme introduced

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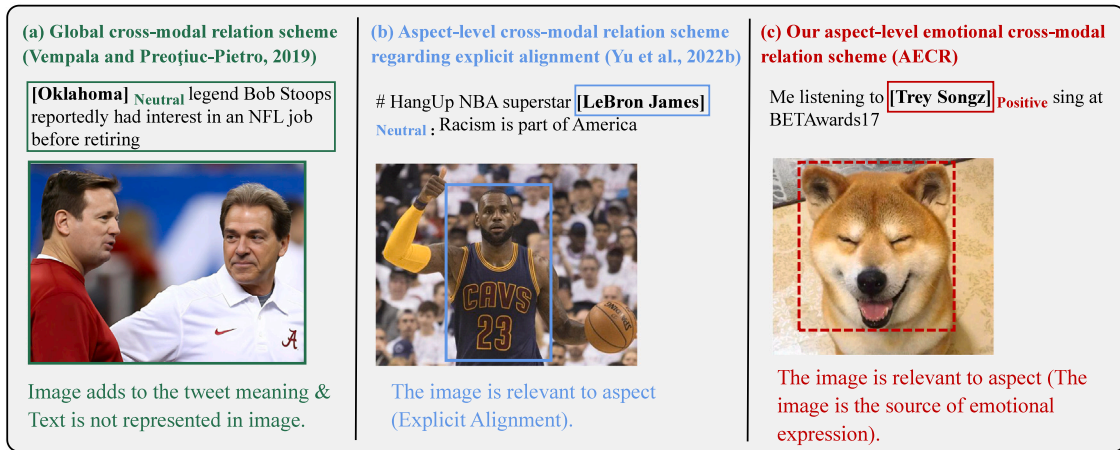


Fig. 1. Examples of our cross-modal relevance definition versus Vempala (Vempala & Preotiu-Pietro, 2019) and Yu’s (Yu et al., 2022) relevance definition. The solid box in the image denotes an explicit alignment, whereas the dashed box represents an implicit emotional connection. Correspondingly, the solid box in the text highlights the textual content associated with the solid or dashed box in the image.

by Vempala and Preotiu-Pietro (2019) and achieve improved performance. In their cross-modal relation scheme, they consider the relation between the entire text and the image from a global perspective. When the image intuitively describes or complements the content of the text, it is regarded as relevant to the overall text. Their annotation scheme has proven to be beneficial for MASC (Ju et al., 2021), however, it still has certain limitations in effectively assisting MASC. MASC task primarily focuses on the sentiment associated with each aspect in the text. In the texts containing multiple aspects, the relevance between individual aspects and the image does not always align with the overall global image–text relevance. Consequently, relying solely on global image–text relevance can introduce irrelevant visual information, ultimately leading to inaccurate sentiment judgments for specific aspects. For example, in Fig. 1-(a), although the image is relevant to the overall text, the aspect “Oklahoma” is irrelevant to the image. Depending on global image–text relevance in this case results in an incorrect sentiment prediction. To capture fine-grained aspect-level cross-modal relevance, Yu, Wang, Xia, and Li (2022) propose an aspect-image relevance dataset. In their relation scheme, the relevance between aspects in the text and image is defined based on **explicit alignment**, which means the image explicitly depicts or includes the aspect mentioned in the text. Specifically, if the aspect is explicitly aligned with the image, it is considered relevant. As shown in Fig. 1-(b), the aspect “LeBron James” is clearly depicted in the image, indicating that this aspect is relevant to the image. Although both global cross-modal relevance and aspect-level cross-modal relevance improve the performance of MASC, these approaches still overlook the **implicit emotional connections**. Such connections arise when the aspect is not explicitly present in the image, yet the image still enhances or conveys the sentiment toward the aspect mentioned in the text. For instance, in Fig. 1-(c), the aspect “Trey Songz” does not appear explicitly in the image, yet the dog’s cheerful expression conveys a positive sentiment to the aspect.

To address the above limitations, we propose an aspect-level emotional cross-modal relation scheme (AECR) for aspect-level image–text relevance, which captures both explicit alignments and implicit emotional connections between aspects in text and image. As illustrated in Fig. 1, although the aspect “Trey Songz” is not explicitly shown in the image, the user expresses a positive sentiment polarity toward “Trey Songz” through the happy expression of the dog in the image. Therefore, in our cross-modal relation scheme, there exists an implicit emotional connection between the image and the aspect. Specifically, the image serves as a source of emotion for the entire image–text pair. Therefore, we consider the image and text to be relevant. Based on our cross-modal relation scheme, we construct a new Aspect-level Emotional Cross-modal Relation dataset, namely AECR-Twitter. To validate the effectiveness of the proposed cross-modal relation scheme and the newly constructed AECR-Twitter dataset in advancing the downstream MASC task, we explore and compare various approaches for integrating cross-modal relevance information into MASC, including lightweight multi-modal models and large language models (LLMs). Extensive experiments are conducted on two widely used datasets, Twitter-15 and Twitter-17. Concretely, our key contributions can be summarized as follows:

- We propose an aspect-level emotional cross-modal relation scheme to enhance MASC by capturing both explicit alignments and implicit emotional connections between text and image. Based on this scheme, we manually construct a high-quality cross-modal relation dataset, containing 3562 text–image pairs from twitter.
- We explore and compare various approaches to leverage cross-modal relevance for MASC, including multi-modal models and LLMs. Specifically, for multi-modal models, we develop an additional cross-modal relation inference module to incorporate cross-modal relevance, while for LLMs, we introduce relation-aware knowledge through prompting, establishing benchmark results under both zero-shot and fine-tuning scenarios.
- We conduct comprehensive experiments on the widely used Twitter-15, Twitter-17, and AECR-Twitter datasets. The results demonstrate that integrating our proposed aspect-level emotional cross-modal relation scheme and dataset leads to consistent performance improvements across different models. Further, we perform in-depth analysis to provide deeper insights into the effectiveness of our method.

We release our AECR-Twitter dataset at <https://github.com/li9527yu/AECR-Twitter>.

2. Research objectives

This paper aims to outline three key research objectives, as detailed below:

- **RO1:** Propose an aspect-level emotional cross-modal relation scheme to capture both explicit alignments and implicit emotional connections between images and text, thereby benefiting MASC. (Section 4.1)
- **RO2:** Construct high-quality aspect-level emotional cross-modal relation datasets to provide data support for facilitating cross-modal relation-aware MASC. (Sections 4.2, 4.3, and 5)
- **RO3:** Incorporate the proposed aspect-level emotional cross-modal relevance into the MASC models to control the contribution of the image. This helps reduce the interference of irrelevant images during multi-modal aspect-based analysis. (Section 6)

3. Related work

3.1. Multi-modal Aspect-based Sentiment Analysis

Multi-modal Aspect-based Sentiment Analysis (MABSA) is a popular field in affective computing. Aiming to analyze users' emotional tendencies toward specific aspects by multi-modal data. Existing work can be categorized into the following types. Firstly, they utilize contextual text features to obtain relevant image features by attention mechanisms (Xu, Mao, & Chen, 2019; Yu & Jiang, 2019; Yu, Jiang, & Xia, 2019). Moreover, some approaches convert images into textual captions to capture the salient features in the images (Wang, Tian et al., 2024; Xiao et al., 2023; Zhao et al., 2022). Recent works attempt to enhance the performance of MABSA by leveraging the cross-modal relation (Ju et al., 2021; Yu et al., 2022; Zhou et al., 2023). Ju et al. (2021) design a cross-modal relation detection module that considers whether the image provides additional information. Compared with Ju et al. (2021) and Yu et al. (2022) utilizes the coarse and fine-grained matching to determine the relevance between aspects and images. Furthermore, recent research leverages the powerful cross-modal capabilities of large vision-language models (LVLMs) to perform instruction tuning or generate context enriched with world knowledge (Feng, Lin, Shang, & Gao, 2024; Wang, Ding et al., 2024). In addition, researchers explore novel approaches to cross-modal alignment. For instance, Luwei, Mao, Zhang, He, and Cambria (2024) are the first to incorporate the aesthetic attributes, analyzing implicit emotional cues in images. Meanwhile, Chen et al. (2024) apply optimal transport theory to design an effective mechanism for filtering out aspect-irrelevant visual representations. Zhou et al. (2023) propose an aspect-oriented method to detect aspect-relevant semantic and sentiment information.

However, existing methods only consider the explicit alignment between aspects and objects in images or the global relevance of text-image pairs, overlooking the role of implicit emotional relation. Therefore, we define an aspect-level cross-modal relation scheme from an emotional perspective. Incorporating this emotional semantic-aware relevance helps reduce interference from irrelevant modality.

3.2. Cross-modal relation inference

Cross-modal relation modeling serves as a fundamental part in many multi-modal applications. Early cross-modal relation is defined for specific domains such as comic books, web pages, and scientific articles (Marsh & Domas White, 2003). Recently, researchers focus on defining cross-modal relevance for open-source multi-modal data, such as social media. Chen, Lu, Kan, and Cui (2013) define cross-modal relation based on the visual relevance between images and text. Other studies define relevance based on the semantic relationship between images and text (Vempala & Preoțiu-Pietro, 2019; Wang et al., 2014). Vempala and Preoțiu-Pietro (2019) build the relevance between text and image based on whether the text or the image provides additional information. Recent studies attempt to explore the implicit or emotional-semantic relevance between images and text (Chen, Su, Wu, & Hua, 2023; Xu, Wang, Tian, Zhang, & Mao, 2022). Chen et al. (2023) define relevance by considering the degree of emotional-semantic alignment between the entire image and text. Additionally, limited works is focusing on the relation between aspects and images. Existing definitions of aspect-level cross-modal relevance considers the explicit alignment (Yu et al., 2022).

However, they overlook implicit relevance, where the aspect is not present in the image but remains emotionally relevant. Therefore, in this work, we propose an emotional semantic-aware aspect-level cross-modal relation scheme that includes both explicit and implicit aspect-level cross-modal relevance.

4. Aspect-level emotional cross-modal relation annotation

In this section, we provide a detailed description of the annotation methodology and process for the construction of AECR-Twitter dataset.

Happy birthday [Garrett Walker]**Positive** !



(a) The image is semantically related to the text but emotionally irrelevant

[SamHunt]**Positive** Performs at Stagecoach.



(b) The image is the source of emotional expression

[Justin Bieber]**Negative** broke history and global records but he only won 2 awards.



(c) The image enhances the emotional expression conveyed in the text

Warriors overwhelm [TrailBlazers]**Negative** 110-99 go up 2-0 in series # NBAPlayoffs



(d) The image is irrelevant to aspect

Fig. 2. Examples of four types of text-image relation in this work.

4.1. AECR-Twitter annotation scheme

After a thorough investigation of previous works on cross-modal relation annotations (Chen et al., 2013, 2023; Vempala & Preoȃuc-Pietro, 2019; Xu et al., 2022), and taking into account the characteristics of the MASC task, we propose an emotional semantic-aware aspect-level text-image relation annotation scheme. This scheme classifies text-image pair relations as either relevant or irrelevant, based on their explicitly expressed objects and implicitly conveyed emotional semantic information. Building on this scheme, we develop systematic annotation guidelines with detailed illustrations and examples to construct high-quality aspect-level emotional cross-modal relation data that supports downstream aspect-based sentiment analysis. Specifically, for the text-image relevant relation, there includes three types:

- **The image is semantically related to the text but emotionally irrelevant (Semantically related but emotionally irrelevant).** The image explicitly contains the aspect mentioned in the text but does not provide any relevant emotional information. Fig. 2-(a) provides an example where the image contains the aspect, specifically the person named Garrett Walker, but does not convey any emotional information related to the tweet.
- **The image is the source of emotional expression (Emotional expression source).** The emotional semantic information in the image is the basis for determining the overall multi-modal sentiment, while the text alone cannot provide this determination. For example, in Fig. 2-(b), the man's expression in the image supplements the emotional expression, clarifying the ambiguous emotional tendency toward aspect "SamHunt" in the text, thereby serving as the emotional source for the aspect.
- **The image enhances the emotional expression conveyed in the text (Enhancing emotional expression).** The image enhances the emotional expression of the aspect by providing additional emotional semantic information. For example, in Fig. 2-(c), the woman's sad expression in the image reinforces the disappointment emotion toward "Justin Bieber" only winning two awards, illustrating how images can enhance emotional expression.

For the text-image irrelevant relation, it occurs when the text-image pair does not belong to any of the above 3 types, which means the image neither contains the aspect mentioned in the text nor provides any additional emotionally relevant information. For example, in Fig. 2-(d), the aspect "TrailBlazers" refers to the NBA team; however, the image presents the NBA team "Warriors"

Table 1
Label distribution in our AECR-Twitter dataset.

Relevance	Type	Number	Proportion
Relevant	The image is semantically related to the text but emotionally irrelevant	868	24.37%
	The image enhances the emotional expression conveyed in the text	730	20.49%
	The image is the source of emotional expression	248	6.96%
Irrelevant	The image is irrelevant to aspect	1716	48.18%

and is therefore clearly irrelevant to the aspect “TrailBlazers”. Considering the image’s content in this case would only introduce irrelevant information.

Unlike previous annotation schemes, which typically define relevant text–image relations only when the aspect in the text explicitly aligns with the object in the image (Yu et al., 2022), dismissing other cases as irrelevant, or focus on the global relevance between the entire text and image (Chen et al., 2023; Vempala & Preoȃuc-Pietro, 2019) rather than aspect-based relevance, our approach accounts for both explicit aspect-based alignment and implicit emotional-semantic relations between the text and image. This means that even if the aspect is not present in the image, we still consider the text–image pair relevant if both contribute to the overall multi-modal sentiment expression. For instance, in Fig. 2-(c), there is no explicit alignment between the aspect “Justin Bieber” in the text and the object in the image. However, the image reinforces the feelings of regret and sadness expressed toward “Justin Bieber”. Recognizing this implicit emotional relevance, rather than dismissing it as irrelevant, allows us to better leverage previously overlooked but valuable text–image relations, ultimately improving aspect-based sentiment analysis.

4.2. Data selection

Our aspect-level emotional cross-modal relation dataset is based on Twitter-17 (Yu & Jiang, 2019), following prior works (Chen et al., 2023; Yu et al., 2022). Twitter-17 consists of multi-modal posts collected from the social media platform Twitter, including text–image pairs annotated with textual aspects and corresponding sentiment for each aspect. We select the Twitter-17 training set which contains 3562 samples to annotate the sentiment label of aspects in each unimodal context and the aspect-level emotional cross-modal relation. The sentiment annotations of aspects under unimodal contexts serve as auxiliary signals to facilitate more accurate labeling of aspect-level emotional cross-modal relation types. We then randomly split the AECR-Twitter dataset into training, development, and test sets with an 8:1:1 ratio, with detailed statistics provided in Table 3. Importantly, the dataset has been carefully curated to exclude harmful content.

4.3. Quality control

We employ 4 postgraduate students as annotators and select 1 experienced annotator as an expert to resolve any annotation inconsistencies. We compensate the annotators according to both the quality and quantity of their annotations. Annotators are asked to annotate the aspect-level emotional cross-modal relation concerning our compiled annotation guidelines. For each tweet’s text–image pair, the annotation process consists of three steps. First, we present the tweet text and aspect entity, asking the annotator to determine the sentiment of the aspect based solely on the text. Second, we provide only the image and aspect entity, asking the annotator to determine the sentiment from the visual content. Finally, we present the text–image pair along with the multi-modal aspect-based sentiment label from Twitter-17 for reference, and ask the annotators to identify the aspect-level emotional cross-modal relation type by fully considering both this reference and their annotations from the previous two steps.

Before starting the formal annotation process, each annotator undergoes several hours of training to be familiar with the annotation guidelines and our developed annotation tool. During annotation, we employ a strict double-annotation strategy to ensure data quality. Each task is randomly assigned to two annotators who independently provide their annotations. If their results are consistent, the annotation is accepted as final; if not, a third expert will make the final decision.

5. Statistics and analysis of AECR-Twitter dataset

In this section, we analyze the newly constructed dataset to gain more insights on the aspect-level emotional cross-modal relation.

Inter-annotator Consistency. As aforementioned, each task is assigned to two annotators for independent annotation. We calculate the inter-annotator consistency ratio for the annotated cross-modal relations and find that 84% of the tasks yield the same results from both annotators, while 16% require further review by a third expert annotator. Through our investigation, we find that the main cause of annotation inconsistencies arises from differences in annotators’ interpretation of emotional expressions in images and text. This suggests that performing strict double annotation is essential to ensure data quality.

Label Distribution. Table 1 presents the label distribution of our dataset. “Irrelevant” label accounts for a proportion of 48.18%. This reflects the nature of social media platforms, and there are two main reasons for this high proportion. First, in the tweet that contains multiple aspects simultaneously, the image is typically related to only one aspect, while being irrelevant to the others. Second, users may also prefer to attach images that are entirely unrelated to the text when posting tweets. For the “Relevant” label, the most common type is that the image contains the aspect within the text but does not convey any emotional information, accounting for 24.37%. The second most frequent type occurs when the image enhances the emotional expression by providing

Table 2
Comparison of relevance distribution between ours and previous annotations.

	Relevant	Irrelevant
Yu et al. (2022)	39.1%	60.9%
Our annotation	45.3% (Explicit: 37.7% Implicit: 7.6%)	54.7%

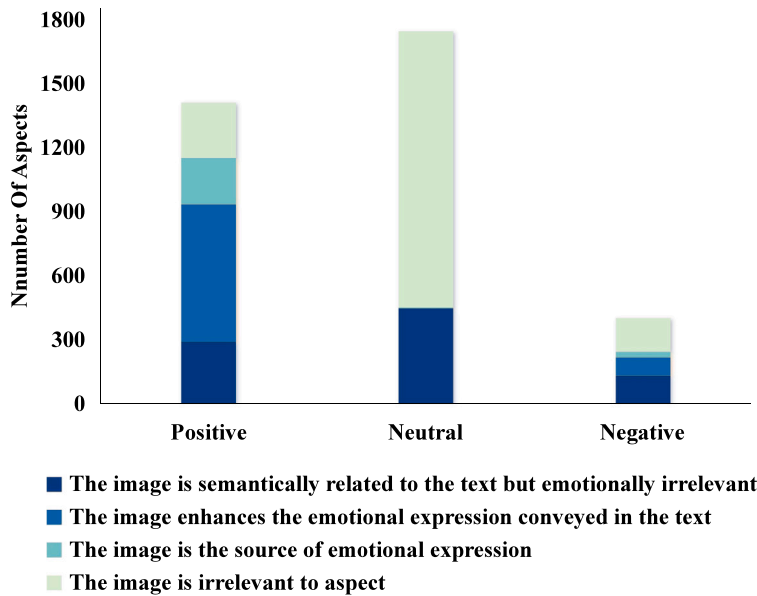


Fig. 3. Relationship between text-image relevance and sentiment in our AECR-Twitter dataset.

additional emotional information. The least common type is when the sentiment of the tweet is conveyed solely through the image. This suggests that users prefer to use text as the primary modality for expressing emotions, with images typically serving as a complement to the text. It is less common for emotions to be conveyed solely through images.

Compared with previous annotations. We compare the distribution of relevant and irrelevant labels in our dataset with that in previous work and present the proportions of explicit and implicit relevance, as shown in Table 2. Compared to Yu et al. (2022), which considers only text-image pairs with explicit alignments as relevant while overlooking implicit emotional connections, our dataset has a higher proportion of relevant labels due to the inclusion of implicit cross-modal relevance. Among the relevance types we annotated, explicit and implicit relevance account for 37.7% and 7.6%, respectively. This suggests that in image-text-related scenarios, approximately one-sixth of emotional expressions are conveyed through implicitly emotion-associated images, such as memes. Compared to previous cross-modal relation annotation schemes, our approach effectively captures these implicit emotional connections between images and text, thereby better supporting downstream MASC.

Relevance Types Regarding Sentiment. In this section, we analyze the association between relevance types and sentiment categories. As shown in Fig. 3, when the sentiment is neutral, the highest proportion of cases occurs when the aspect is unrelated to the image, followed by cases where the image and aspect are semantically related but not emotionally connected. This suggests that when users have no emotional tendency toward an aspect entity, the image is often unrelated to the aspect or serves merely as a descriptive supplement. For positive sentiment, the highest proportion of cases occurs when the image enhances the emotional expression of the aspect, indicating that users tend to reinforce positive emotions through both text and images. In cases of negative sentiment, the proportions of instances where the image is unrelated, semantically related but emotionally irrelevant, or where the image enhances emotional expression are similar. This suggests that negative emotions are expressed in more diverse ways.

6. Leveraging aspect-level emotional cross-modal relevance for MASC

Our method consists of three main parts. First, we introduce the task formulation, which defines the aspect-based cross-modal relation inference task, the multi-modal aspect-based sentiment classification (MASC) task, and the relationship between them. Second, we present the general architecture commonly adopted in existing MASC models. Finally, we describe our proposed modules designed to incorporate aspect-level emotional cross-modal relevance.

Table 3
Statistics for Twitter-15, Twitter-17 and AECR-Twitter.

Dataset	Twitter-15			Twitter-17			AECR-Twitter		
	Train	Dev	Test	Train	Dev	Test	Train	Dev	Test
#Pos	928	303	317	1508	515	493	1188	150	170
#Neu	1883	149	113	1638	144	168	1327	158	153
#Neg	368	670	607	416	517	336	383	47	33
#Total	3179	1122	1037	3562	1176	1234	2851	355	356
#Sent	2101	727	674	1746	577	587	1672	332	335

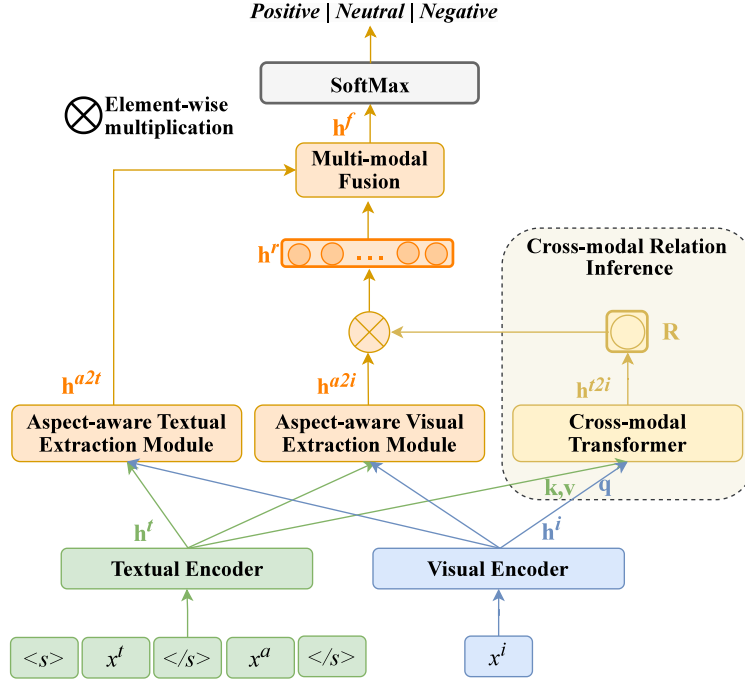


Fig. 4. Overview of EmoCRel-MASC architecture for MASC.

6.1. Task formulation

To enhance sentiment classification performance by reducing the influence of irrelevant images, we incorporate the proposed aspect-level emotional cross-modal relevance into the MASC task. Specifically, the aspect-based cross-modal relation inference task aims to predict the relevance between an image and a given aspect in the text. The relevance probability is then leveraged to improve the multi-modal aspect-based sentiment classification task by suppressing the influence of irrelevant images.

Formally, each sample $\mathbf{x}$ in the corpus consists of a tweet with a text–image pair and its corresponding aspect, where x^t , x^i , and x^a denote the text, image, and aspect, respectively. The text x^t is a sentence composed of n words, while the aspect x^a refers to a named entity, such as a person, location, or organization, and is a sub-sequence of x^t containing m words.

Aspect-based Cross-modal Relation Inference is formulated as a binary classification problem. Given an input $\mathbf{x} = (x^t, x^i, x^a)$, the goal is to predict the relation r between the image x^i and the text x^t with respect to the given aspect x^a , where $r \in \{\text{relevant}, \text{irrelevant}\}$.

Multi-modal Aspect-based Sentiment Classification aims to predict the sentiment polarity label $y \in \{\text{positive}, \text{neutral}, \text{negative}\}$ for a given text x^t and image x^i pair concerning the aspect x^a .

6.2. Basic MASC model architecture

Basic MASC Model Architecture comprises three main components. The Textual/Visual Encoder encodes the text and image into feature representations. The Aspect-aware Extraction Module then extracts the representations that focus on aspect-relevant representations in both modalities. And the sentiment classification module fuses textual and visual features to obtain multi-modal representations, which are fed into a Softmax layer for sentiment prediction.

Textual/Visual Encoder. Given an input $\mathbf{x} = (x^t, x^i, x^a)$, which consists of a text–image pair and an aspect, we use a textual encoder and a visual encoder to extract contextualized feature representations for the textual and visual modalities, respectively. For

the textual encoder, we employ RoBERTa (Liu, 2019) or BERT (Devlin, Chang, Lee, & Toutanova, 2019) to encode the concatenation of text x^t and aspect x^a . For the visual encoder, we utilize a Faster R-CNN (Ren, He, Girshick, & Sun, 2016) or ViT (Dosovitskiy et al., 2020). The resulting representations are denoted as $\mathbf{h}^t$ and $\mathbf{h}^i$ for text and image.

Aspect-aware Extraction Module. After obtaining the text and image representations, we feed them into the aspect-aware extraction module. This module is designed to extract and refine aspect-relevant representations from both text and images. This module can be implemented using various methods, including modules from frameworks such as ITM (Yu et al., 2022), HIMT (Yu, Chen, & Xia, 2023), and DPFN (Wang, Tian et al., 2024). For aspect-aware textual representations, methods like attention mechanisms or dependency parsers are applied to derive the aspect-aware textual representation $\mathbf{h}^{a2t}$. For aspect-aware visual representations, methods such as cross-modal transformer (CMT) (Tsai et al., 2019) are applied to capture the aspect-aware visual representation $\mathbf{h}^{a2i}$.

Sentiment Classification Module. We typically adopt concatenation method in Wang, Tian et al. (2024) or the transformer method in Yu et al. (2023, 2022) to generate the final multi-modal fusion representation $\mathbf{h}^f$. We then feed $\mathbf{h}^f$ into a Softmax layer to predict the probability distribution over sentiment labels $\mathbf{y}$ for sentiment classification:

$$p(\mathbf{y}) = \text{Softmax}(\mathbf{W}_s \mathbf{h}^f + \mathbf{b}_s) \quad (1)$$

During the training process, for each input $\mathbf{x}$, we use the cross-entropy loss to maximize the probability of the gold sentiment label y^* , and the objective function is:

$$\mathcal{L}_s = -\log P(y^*) \quad (2)$$

6.3. Enhancing sentiment prediction with aspect-level emotional cross-modal relation

We propose a **Emotional Cross-modal Relation-aware MASC (EmoCRel-MASC)** to effectively integrate the predicted relevance into MASC in Fig. 4. The cross-modal relation inference module outputs a relevance probability that controls the contribution of the image, and we employ the AECR-Twitter dataset as supervision to train this module. The relation-aware multi-modal fusion module then uses the predicted relevance probability to guide the fusion of textual and visual features, effectively suppressing the influence of irrelevant images.

Aspect-level Emotional Cross-modal Relation Inference. To incorporate our aspect-level emotional cross-modal relation into the MASC task, we employ a cross-modal relation inference module to perform aspect-based cross-modal relation inference and obtain the corresponding relevance probability. This probability controls the contribution of the image and reduces the interference of irrelevant images. Specifically, we first apply a CMT to capture the cross-modal image representation, with the image representation $\mathbf{h}^i$ as query $\mathbf{q}$, and the text representation $\mathbf{h}^t$ as key $\mathbf{k}$ and value $\mathbf{v}$. Then, a max-pooling operation is performed to obtain the most salient cross-modal features $\mathbf{h}^{t2i}$ for cross-modal relation inference:

$$\mathbf{h}^{t2i} = \text{maxpool}(\text{CMT}(\mathbf{h}^i, \mathbf{h}^t, \mathbf{h}^t)) \quad (3)$$

The final relevance probability is computed as follows. Notably, we take the probability of the relevant class predicted by this module as the relevance probability.

$$p(\mathbf{r}) = \text{Sigmoid}(\mathbf{W}_r \mathbf{h}^{t2i} + \mathbf{b}_r) \quad (4)$$

Relation-aware Cross-modal Fusion. To alleviate the interference of irrelevant images in MASC, we design a relation-aware multi-modal fusion module that leverages the relevance probability to guide the model in focusing on relevant images while minimizing the influence of irrelevant ones.

Specifically, we first utilize an aspect-aware visual extraction module to obtain the aspect-aware visual representation $\mathbf{h}^{a2i}$. We then broadcast the cross-modal relevance probability $p(\mathbf{r})$ to match the size of $\mathbf{h}^{a2i}$, resulting in $\mathbf{R}$. This matrix $\mathbf{R}$ is used as weights to obtain the relation-aware cross-modal representation $\mathbf{h}^r$ through element-wise multiplication:

$$\mathbf{h}^r = \mathbf{R} \odot \mathbf{h}^{a2i} \quad (5)$$

For example, when the relevance probability $\mathbf{R}$ approaches 0, indicating that the image and text are irrelevant, the model correspondingly reduces the contribution of visual features during multi-modal fusion. Finally, we input the relation-aware cross-modal representation $\mathbf{h}^r$ and the aspect-aware textual representation $\mathbf{h}^{a2t}$ into the multi-modal fusion module mentioned above to generate the enhanced multi-modal fusion representation $\mathbf{h}^{AECR-f}$. We then use $\mathbf{h}^{AECR-f}$ as the final multi-modal feature input to the Softmax layer for sentiment classification. Moreover, for each input $\mathbf{x}$, we use two separate cross-entropy loss to maximize the probability of its gold cross-modal relation label r^* and the gold sentiment label y^* , and the combined objective function is:

$$\mathcal{L} = \mathcal{L}_s + \mathcal{L}_r = -\log P(y^*) - \log P(r^*) \quad (6)$$

During inference, we select the label with the highest score as our predicted sentiment label.

7. Experiments

Datasets. We conduct experiments on two widely-used benchmark datasets Twitter-15 and Twitter-17 (Yu & Jiang, 2019), as well as our constructed AECR-Twitter dataset, with detailed statistics provided in Table 3.

Implementation Detail. For the backbone models ITM, HIMT, DPFN, A2II, AoM and DQPSA, we follow their original settings. For incorporating our relevance, we employ our AECR-Twitter dataset to train the cross-modal relation inference module. This module provides relation-aware knowledge for the backbone models, guiding the multi-modal fusion process and helping to mitigate the impact of irrelevant images. In Qwen2-VL, we use two separate Qwen2-VL-7B models and apply LoRA for fine-tuning with a learning rate of $1e-4$. For incorporating our relevance, the first Qwen2-VL is fine-tuned on the AECR-Twitter dataset. It is then used to generate predicted relevance labels for Twitter-15 and Twitter-17. The second is fine-tuned for MASC, where the predicted relevance is included in the instruction to guide the model to suppress irrelevant images. For Qwen2.5-VL, we utilize the Qwen2.5-VL-7B model and employ the same experimental setup as Qwen2-VL. All models are implemented using the PyTorch framework and executed on an NVIDIA Tesla V100 GPU. We conduct experiments using three different random seeds and report the average performance. We adopt Dan Bikel's randomized evaluation comparator for significance test (Lenth, 1990).

Evaluation Metrics. Following previous works, we use Accuracy (Acc) and Macro-F1 score (F1) to measure the MASC performance.

Compared Models. To validate the effectiveness of our proposed emotional semantic-aware aspect-level cross-modal relation in advancing MASC, we compare several strong methods with their counterparts enhanced by our cross-modal relation inference module.

- ITM (Yu et al., 2022): a multi-task learning framework that can perform both coarse-grained and fine-grained image-aspect alignment. Since ITM originally utilizes the aspect-image relevance dataset proposed by Yu et al. (2022) to perform aspect-based cross-modal relation inference, We replace Yu's dataset with our AECR-Twitter dataset to train the cross-modal relation inference module in "ITM + Our AECR".
- HIMT (Yu et al., 2023): A hierarchical interactive multi-modal transformer model. In "HIMT+Our AECR", we incorporate the cross-modal relation inference module into HIMT to obtain the relevance weights between the image and the aspect.
- DPFN (Wang, Tian et al., 2024): A dual-perspective fusion network considering both global and local fine-grained sentiment information related to specific aspects. In "DPFN+Our AECR", we apply the same way as in "HIMT+Our AECR" to incorporate our relevance.
- A2II (Feng et al., 2024): A generative model which leverages Language-Vision Large Models (LVLMs) to address the limitations of multi-modal fusion and mitigate the impact of irrelevant image noise through instruction tuning. In "A2II+Our AECR", we use the relation-aware knowledge generated by Qwen2-VL-7B model fine-tuned on the AECR-Twitter dataset to replace that generated by their InstructBlip-Flan-T5-xl model (Dai et al., 2023).
- Qwen2-VL (Wang, Bai et al., 2024): The widely popular LVLMs, which demonstrate satisfactory performance. "Qwen2-VL+Our AECR" refers to injecting our relation-aware knowledge into the instructions.
- Qwen2.5-VL (Bai et al., 2025): An updated version of Qwen2-VL with enhanced visual-language alignment and reasoning capabilities. "Qwen2.5-VL+Our AECR" follows the same relation-aware knowledge injection strategy as "Qwen2-VL+Our AECR".
- AoM (Zhou et al., 2023): An aspect-oriented method which utilizes Graph Convolutional Network (GCN) to fuse semantic and sentiment features guided by syntactic dependencies. "AoM+Our AECR" refers to incorporating our relation-aware knowledge into AoM, following the same strategy employed in "A2II + Our AECR".
- DQPSA (Peng et al., 2024): A multi-modal sentiment analysis framework that uses dual-query prompts for visual-text alignment and an energy-based module for span boundary pairing. In "DQPSA + Our AECR", we incorporate our proposed relation-aware knowledge into DQPSA using the same way as applied in "A2II + Our AECR".

7.1. Main results

Table 4 shows the main results of MASC on Twitter-15, Twitter-17 and AECR-Twitter, where the results for ITM, HIMT, DPFN, A2II, AoM and DQPSA are re-implemented by us. † indicates that the difference between the corresponding model and the model with Our AECR is significant at a confidence level of $p < 0.01$. ◊ means the difference is not significant ($p > 0.05$). We observe consistent improvements across various MASC model frameworks after incorporating our proposed emotion-aware cross-modal relevance. Notably, it also leads to consistent gains in two state-of-the-art methods, AoM and DQPSA, demonstrating the effectiveness of our proposed relevance. The improvement is attributed to the combined effect of our cross-modal relation inference module and our proposed relevance. The module effectively assists existing MASC methods in reducing the impact of irrelevant images, thereby enhancing sentiment analysis performance. Meanwhile, our proposed relevance captures both explicit alignments and implicit emotional connections between textual aspects and image, enabling more comprehensive capture of the relevance between aspects and images.

Qwen2.5-VL and DQPSA achieve superior performance compared to other models, which can be attributed to the strong multi-modal understanding capabilities of LVLMs and the effectiveness of vision-language pre-training tasks. The integration of our proposed cross-modal relevance further boosts the performance of both Qwen2.5-VL and DQPSA, validating the effectiveness of

Table 4

Main results of different approaches on Twitter-15, Twitter-17 and AECR-Twitter.

Methods	Twitter-15		Twitter-17		AECR-Twitter	
	ACC	F1	ACC	F1	ACC	F1
ITM	77.78	73.44	71.64	70.27	71.72	67.23
ITM+Our AECR	78.36 [†] _{±0.20}	74.09 [†] _{±0.11}	72.93 [†] _{±0.91}	71.86 [†] _{±0.76}	73.31 [†] _{±0.39}	69.17 [†] _{±0.55}
HIMT	76.01	71.46	67.77	65.34	68.31	62.05
HIMT+Our AECR	76.37 [°] _{±0.43}	72.01 [†] _{±0.20}	68.82 [†] _{±0.12}	66.88 [†] _{±0.18}	69.94 [†] _{±1.18}	63.88 [†] _{±0.12}
DPFN	76.75	71.72	70.06	68.68	66.29	65.96
DPFN+Our AECR	76.53 [°] _{±0.31}	72.46 [†] _{±0.37}	71.12 [†] _{±0.40}	69.38 [†] _{±0.32}	71.06 [†] _{±1.20}	70.85 [†] _{±1.21}
A2II	76.61	72.70	72.96	71.37	71.15	66.96
A2II+Our AECR	77.77 [†] _{±0.07}	73.13 [†] _{±0.35}	74.11 [†] _{±0.38}	72.74 [†] _{±0.38}	72.19 [†] _{±0.48}	68.63 [†] _{±0.67}
Qwen2-VL	77.91	73.41	72.69	71.07	71.91	67.12
Qwen2-VL+Our AECR	79.36 [†] _{±0.72}	74.25 [†] _{±0.36}	73.39 [†] _{±0.12}	72.61 [†] _{±0.29}	73.68 [†] _{±1.13}	70.06 [†] _{±1.13}
Qwen2.5-VL	77.62	73.73	74.31	73.95	72.75	68.30
Qwen2.5-VL+Our AECR	78.73 [†] _{±0.07}	74.19 [†] _{±0.07}	74.55[†] _{±0.11}	74.40[†] _{±0.36}	73.87 [†] _{±0.28}	71.10 [†] _{±0.02}
AoM	77.90	73.84	73.52	72.38	73.37	70.17
AoM+Our AECR	78.52 [†] _{±0.23}	75.08 [†] _{±0.25}	74.02 [†] _{±0.09}	72.92 [†] _{±0.31}	74.15[†] _{±0.24}	72.05[†] _{±0.40}
DQPSA	80.80	80.80	71.88	71.88	71.34	71.34
DQPSA+Our AECR	81.65[†] _{±0.20}	81.65[†] _{±0.20}	72.89 [†] _{±0.29}	72.89 [†] _{±0.29}	72.28 [†] _{±0.42}	72.28 [†] _{±0.42}

Table 5

Results of integrating cross-modal relevance into zero-shot LLMs on Twitter-15 and Twitter-17.

Methods	Twitter-15		Twitter-17	
	ACC	F1	ACC	F1
GPT-3.5-Turbo	48.72	49.72	51.95	48.31
GPT-3.5-Turbo+Our AECR	52.49	52.36	52.55	49.25
Qwen-Turbo	65.19	59.80	57.37	54.57
Qwen-Turbo+Our AECR	68.56	61.83	59.72	56.02
DeepSeek-V3	62.64	61.64	59.40	58.87
DeepSeek-V3+Our AECR	66.12	63.67	62.53	61.29

our emotion-aware cross-modal relevance in enhancing MASC. Subsequently, DQPSA outperforms Qwen2.5-VL on the Twitter-15 dataset. We believe this is due to its use of prompt-controlled image comprehension during the pre-training phase, which enables the model to focus more precisely on image regions relevant to the given aspect, leading to more accurate sentiment predictions. In contrast, “Qwen2.5-VL+Our AECR” achieves better results on the Twitter17, owing to its superior generalization ability. Nevertheless, on the AECR-Twitter dataset, Qwen2.5-VL still performs slightly worse than AoM and DQPSA, possibly because Qwen2.5-VL require more data to fully adapt, while AoM and DQPSA are better suited to small-scale datasets.

Overall, our proposed emotion-aware cross-modal relevance yields consistent improvements across all model architectures, highlighting its robustness and effectiveness in capturing implicit emotional associations and enhancing sentiment analysis performance.

7.2. Performance of integrating cross-modal relevance into zero-shot LLMs

We further investigate the effectiveness of our aspect-level emotional cross-modal relation in enhancing the capabilities of various LLMs through prompting in zero-shot scenarios. Fig. 5 shows our designed prompt template. Specifically, the task definition outlines the MASC task and explains how cross-modal relevance is leveraged for MASC. We provide the aspect, sentiment options, and the sentence in tweet in the prompt. The image descriptions are extracted from BLIP-Image and BLIP-VQA (Li, Li, Xiong, & Hoi, 2022) following Yang, Wang, Li, Na, and Yu (2024), and consist of image captions, visual entity description, and visual sentiment descriptions.

For the cross-modal relation, we fine-tune the Qwen2-VL-7B model on the AECR-Twitter dataset to generate cross-modal relation-aware knowledge. We evaluate the MASC performance of various open-source and closed-source LLMs, including DeepSeek-V3, GPT-3.5-Turbo, and Qwen-Turbo.

As shown in Table 5, all LLMs exhibit consistent performance improvements when incorporating the cross-modal relation-aware knowledge under zero-shot scenarios. This observation demonstrates that our relation-aware knowledge improves the reliability of LLMs in the MASC task. In particular, open-source LLMs DeepSeek-V3 and Qwen-Turbo outperform the proprietary GPT-3.5-Turbo in both datasets.

7.3. Comparison with previously defined cross-modal relevance

We adopt the following four definitions of cross-modal relations for comparison based on the framework of Yu et al. (2022).

LLM Prompt Template

Task Definition: Detect the sentiment polarity towards the aspect from the sentence. Strictly format output as [sentiment]. If cross-modal relation is relevant, integrate the information of the sentence and the image description to make predictions. if cross-modal relation is irrelevant, only use the sentence to make predictions.

Aspect: Trey Songz

Options: -positive -neutral -negative

Make sentiment prediction for the following sentence:
Sentence: Me listening to Trey Songz sing at BETAwards17.
Image Description : The image is about a dog smiling happily. The image implies happy sentiment. The entity of the dog is present in the image.
Cross-modal relation: relevant

Fig. 5. Relevance-aware zero-shot prompt template.

Table 6
Results of previously defined relevance on Twitter-15 and Twitter-17.

Methods	Twitter-15		Twitter-17	
	ACC	F1	ACC	F1
Baseline (T)	76.28	71.36	69.77	68.00
Baseline (M)	76.60	71.35	70.50	68.85
+ Vempala's rel	77.27	72.33	71.31	69.88
+ Chen's rel	76.88	72.33	71.28	70.05
+ Yu's rel	77.78	73.44	71.64	70.27
+ Our AECR	78.36	74.09	72.93	71.86

- Baseline (T): A text-based method that uses the textual encoder as Yu et al. (2022).
- Baseline (M): A multi-modal method that utilizes the textual and visual encoder as Yu et al. (2022).
- + Yu's rel: We incorporate the cross-modal relation inference module into the Baseline (M), and train this module using the cross-modal relevance data constructed by Yu et al. (2022), which defines relevance based on whether the aspect explicitly appears in the image.
- + Vempala's rel: The architecture is the same as that of "+Yu's rel", with the only difference being the use of the cross-modal relevance proposed by Vempala and Preoȃiuc-Pietro (2019), which considers the global relevance between the entire image and text. In their definition, relevance is determined based on whether the image adds to the meaning of the tweet.
- + Chen's rel: We adopt the same architecture as "+Yu's rel", but train the cross-modal relation inference module with the relevance data proposed by Chen et al. (2023), where the relevance is defined based on the degree of emotional-semantic alignment between the entire image and text.

The result is shown in Table 6. First, comparing "Baseline (T)" and "Baseline (M)" shows that incorporating multi-modal inputs improves aspect-based sentiment classification. This improvement can be attributed to the fact that images enrich textual information with additional emotional and semantic cues. Second, compared to "Baseline (M)", all models that incorporate relevance (rows 3–5 in Table 6) consistently achieve better performance, suggesting that integrating cross-modal relevance helps the model focus on relevant images.

Finally, incorporating our proposed emotion-aware cross-modal relevance clearly yields the best results, highlighting the effectiveness of our method. Unlike previously defined cross-modal relevance, which either focuses solely on the explicit alignment between the aspect and the image object (Yu et al., 2022), or considers global relevance between the entire text and image (Chen et al., 2023; Vempala & Preoȃiuc-Pietro, 2019).

7.4. Impact of leveraging different cross-modal relation types

To better understand the contribution of each relation type as illustrated in Fig. 2, we conduct an ablation study on the AECR-Twitter dataset by omitting one cross-modal relation type each time and compare the results as shown in Table 7. "Complete Model" represents the MASC model that is aware of all four types, and "without x type" indicates disregarding the relation type x when

Table 7
Impact of leveraging different cross-modal relations.

Methods	AECR-Twitter	
	ACC	F1
Complete model	73.31	69.17
Without “Irrelevant” type	63.34	57.69
Without “Emotional expression source” type	72.46	68.44
Without “Enhancing emotional expression” type	71.62	67.36
Without “Semantically related but emotional irrelevant” type	71.20	67.05

Table 8
Impact of different proportions of gold relation label on our AECR-Twitter dataset.

Methods	AECR-Twitter	
	ACC	F1
Without relevance	72.19	65.58
With silver relevance	72.75	66.42
With 50% gold relevance	78.93	70.17
With 100% gold relevance	81.46	71.62

performing MASC. Note that the training set size remains the same across the ablation study. When a relation type is removed, we retain the full dataset for training the MASC task, while masking the supervision signal for that type in the aspect-based cross-modal relation inference task.

Table 7 shows the ablation results. We first observe that the absence of any type leads to consistent performance degradation compared with the “Complete Model”, indicating that all relation types contribute positively to the MASC task. Notably, the removal of “Irrelevant” type causes the most significant performance drop. We attribute this to the prevalent weak image–text alignment in social media, where excluding such samples critically impairs the model’s ability to filter cross-modal noise. Moreover, the elimination of both “Enhancing emotional expression” and “Semantically Related but Emotionally Irrelevant” types results in a substantial decline. This highlights the significant roles of these two types: the former strengthens textual sentiment expression through visual implicit emotional cues, while the latter assists the model in distinguishing between emotional relevance and semantic relevance, effectively capturing crucial semantic information within images. The ablation of “Emotional expression source” type results in minor performance degradation. This potentially stems from the low proportion of this type in the dataset, which prevents the model from adequately learning this type’s patterns.

7.5. Analysis of the impact of relevance on MASC

To evaluate the impact of cross-modal relevance label quality on MASC task performance, we analyze the impact of leveraging silver and gold relevance in our AECR-Twitter dataset. Specifically, we compare the performance of models without leveraging cross-modal relevance, using silver relevance labels, using 50% gold relevance labels, and using 100% gold relevance labels, based on the framework of Yu et al. (2022). Silver relevance labels refer to predicted annotations produced by the Aspect-level Emotional Cross-modal Relation Inference module, which was trained on the AECR-Twitter dataset. As shown in Table 8, all methods incorporating relevance outperform that without relevance in both accuracy and F1 score, indicating that cross-modal relevance significantly aids in establishing connections between different modalities for sentiment classification. Moreover, we observe that as the proportion of gold labels increases, MASC performance shows a marked improvement. This highlights that higher-quality relevance labels further enhance MASC performance.

7.6. Analysis on error patterns

To better understand the improvements achieved by our method, we analyze how the number of error patterns changes between different comparison models under the framework of Yu et al. (2022). An error pattern {X, Y} indicates that the gold sentiment label “X” was incorrectly predicted as “Y” or vice versa. Fig. 6 shows the results on the Twitter-17 dataset.

We observe that among the three error patterns, the “{Neu, Pos}” and “{Neu, Neg}” errors occur most frequently, indicating that distinguishing neutral sentiment from positive or negative sentiment remains a major challenge. These errors often occur when the emotional clues in the image are ambiguous, leading the model to predict a stronger sentiment. By incorporating our relevance, the model can better utilize the implicit emotional cues in images to distinguish between neutral sentiment and the other two sentiment labels. In contrast, the “{Neg, Pos}” error pattern appears least frequently but often involves sarcasm or conflicting multi-modal cues. For instance, a tweet expressing negative sentiment may be accompanied by a positive image, which can mislead the model into making incorrect predictions. Our method mitigates this issue by down-weighting images that are irrelevant to the aspect, allowing the model to focus on the sarcastic textual cue. Although our model performs better than Yu et al. (2022) in handling this type of error, it is still slightly outperformed by the baseline. We will explore appropriate methods to address this type of error in future work.

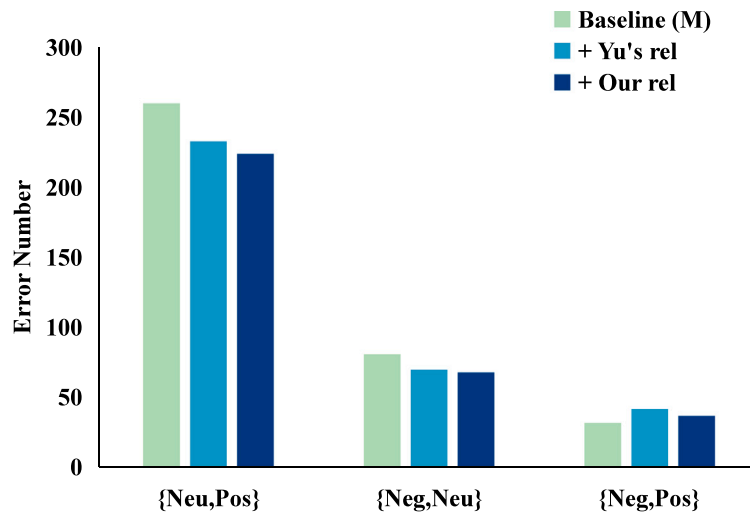


Fig. 6. Statistics of all error patterns. Note that ‘Pos’, ‘Neu’, and ‘Neg’ represent Positive, Neutral, and Negative.

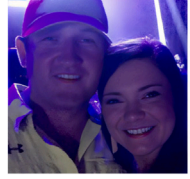
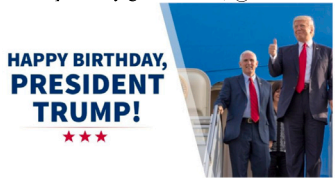
	Gold	Positive
	+ Vempala's rel Pred	Neutral ✗
	+ Chen's rel Pred	Neutral ✗
	+ Yu's rel Pred	Neutral ✗
	+ Our AECR Pred	Positive ✓
	Gold	Neutral
	+ Vempala's rel Pred	Positive ✗
	+ Chen's rel Pred	Positive ✗
	+ Yu's rel Pred	Neutral ✓
	+ Our AECR Pred	Neutral ✓

Fig. 7. Results of different relevance in the two test cases.

7.7. Case study

Fig. 7 presents a case study comparing the performance when integrating our proposed emotion-aware cross-modal relevance with the performance achieved integrating other previously defined cross-modal relevance types, under the MASC model framework of Yu et al. (2022). In Case1, as shown in the left of Fig. 7, given the aspect “bombfactory”, only “+ Our AECR” produces the correct prediction. This is attributed to the distinction in our relevance definition: while the aspect is not explicitly depicted in the image, the image still conveys an implicit emotional connection toward the aspect that our method takes into account. In contrast, previous relevance definitions ignore this difference. Consequently, our relevance allows the model to focus on the smiling expression of people in the image as a source of sentiment, enabling it to make correct predications.

In the Case2, as shown in the right of Fig. 7, given the aspect “United States”, both “+ Vempala’s rel” and “+ Chen’s rel” make incorrect predictions. This is due to they focus on the global relevance between the image and text, mistakenly assuming a connection between the image and the aspect. In contrast, “+ Yu’s rel” and “+ Our AECR” adopt an aspect-level cross-modal relevance approach, correctly identifying that the aspect is unrelated to the image.

8. Conclusions

To fully capture both explicit alignments and implicit emotional connections between text and image to enhance MASC, this paper proposes the aspect-level cross-modal relation scheme and manually constructs the high-quality AECR-Twitter dataset. We provide a comprehensive description of the AECR-Twitter construction process and offer in-depth data analysis. We explore and compare

various methods for integrating cross-modal relevance into MASC and conduct extensive experiments. The results demonstrate the consistent benefits of incorporating our proposed emotion-aware cross-modal relevance for enhancing MASC performance.

Future work will focus on several key areas. First, despite our efforts, the AECR-Twitter dataset remains relatively small. We plan to expand it by collecting more text–image pairs from real-world scenarios and generating additional high-quality labeled AECR-Twitter data to further support MASC. Second, while we have classified relevant text–image pairs into three fine-grained categories based on our annotation guidelines, our current MASC modeling only considers binary relevance (i.e., relevant vs. irrelevant). In future work, we aim to integrate these fine-grained relevance categories to enhance our approach. Third, while our aspect-level emotional cross-modal relation scheme is currently leveraged for MASC, it has potential benefits for several other multi-modal applications as discussed above, we will apply it to more practical applications in the future.

9. Limitations

While our proposed aspect-level emotional cross-modal relation scheme achieves the consistent improvements across multiple model architectures, there are several limitations to our work. The construction of the AECR-Twitter dataset relies on manual annotation, which incurs costs in terms of time, and labor. We plan to extend this dataset to other multi-modal applications in the future. Moreover, our current relation scheme focuses primarily on the social media domain. We aim to expand it to other domains in subsequent work.

CRedit authorship contribution statement

Zhaoyu Li: Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Chen Gong:** Writing – review & editing, Validation, Resources, Funding acquisition, Conceptualization. **Guohong Fu:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

We have published relevant codes and data in an anonymous GitHub repository.

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